

Introduction to Social Data Analytics:

Lecture 9 Part 1

U.S. 2000 Presidential Election

- ▶ George Bush vs. Al Gore
- ▶ Bush: 271 electoral votes, Gore: 266
- ▶ Very close election, long recount

Controversy in Florida

- ▶ Florida, 25 electoral votes, was particularly close
 - ▶ In the end, Bush won Florida by just 537 votes!
- ▶ Of particular controversy: butterfly ballot in Palm Beach County
- ▶ Many voters complained that they had intended to vote for Gore, but had actually voted for Pat Buchanan (Reform Party).

Butterfly Ballot

PALM BEACH COUNTY, FLORIDA NOVEMBER 7, 2000			PALM BEACH COUNTY, FLORIDA NOVEMBER 7, 2000		
es will electors.)	(REPUBLICAN)				
	GEORGE W. BUSH - PRESIDENT	3 ➡			
	DICK CHENEY - VICE PRESIDENT				
	(DEMOCRATIC)		4 ←	(REFORM)	
	AL GORE - PRESIDENT	5 ➡		PAT BUCHANAN - PRESIDENT	
	JOE LIEBERMAN - VICE PRESIDENT			EZOLA FOSTER - VICE PRESIDENT	
	(LIBERTARIAN)		6 ←	(SOCIALIST)	
HARRY BROWNE - PRESIDENT	7 ➡		DAVID McREYNOLDS - PRESIDENT		
ART OLIVIER - VICE PRESIDENT			MARY CAL HOLLIS - VICE PRESIDENT		
(GREEN)		8 ←	(CONSTITUTION)		
RALPH NADER - PRESIDENT	9 ➡		HOWARD PHILLIPS - PRESIDENT		
WINONA LaDUKE - VICE PRESIDENT			J. CURTIS FRAZIER - VICE PRESIDENT		
(SOCIALIST WORKERS)		10 ←	(WORKERS WORLD)		
JAMES HARRIS - PRESIDENT	11 ➡		MONICA MOOREHEAD - PRESIDENT		
MARGARET TROWE - VICE PRESIDENT			GLORIA La RIVA - VICE PRESIDENT		
(NATURAL LAW)					
JOHN HAGELIN - PRESIDENT	13 ➡				
NAT GOLDBABER - VICE PRESIDENT					
			WRITE-IN CANDIDATE To vote for a write-in candidate, follow the directions on the long stub of your ballot card.		

Palm Beach County

- ▶ Wand et al (2001) “The Butterfly Did It: The Aberrant Vote for Buchanan in Palm Beach County, Florida” *American Political Science Review*
- ▶ Use votes from 1996 U.S. Presidential Election to predict those in 2000.
- ▶ Specifically, use votes in each county for Ross Perot (Reform Party) in 1996 to predict votes for Buchanan in 2000.
- ▶ Was Palm Beach County an outlier?

Load the data

```
florida <- read.csv("florida.csv")  
head(florida)
```

```
##      county Clinton96 Dole96 Perot96 Bush00 Gore00 Buchanan00  
## 1  Alachua      40144  25303    8072  34124  47365         263  
## 2   Baker       2273   3684     667   5610   2392          73  
## 3    Bay       17020  28290    5922  38637  18850         248  
## 4 Bradford     3356   4038     819   5414   3075          65  
## 5  Brevard     80416  87980   25249 115185  97318         570  
## 6  Broward    320736 142834   38964 177323 386561         788
```

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.1.2
```

```
## Warning: package 'ggplot2' was built under R version 4.1.2
```

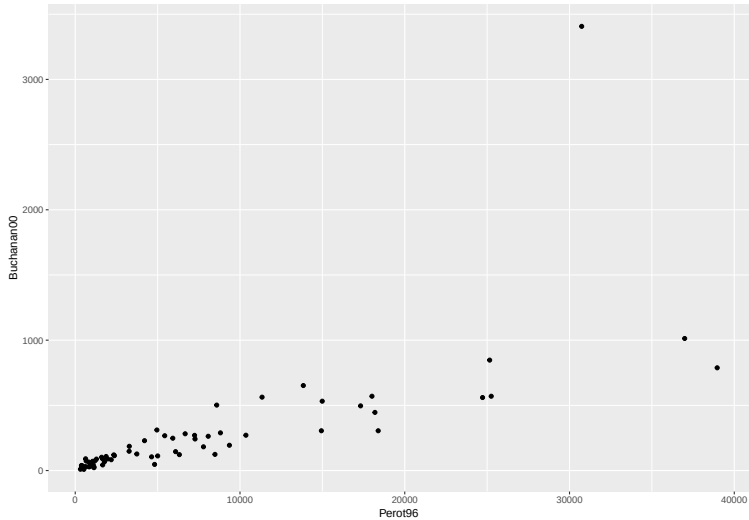
```
## Warning: package 'tibble' was built under R version 4.1.2
```

```
## Warning: package 'tidyr' was built under R version 4.1.2
```

Plot votes for Perot and votes for Buchanan

```
ggplot(florida, mapping=aes(x=Perot96, y=Buchanan00)) +  
  geom_point()
```

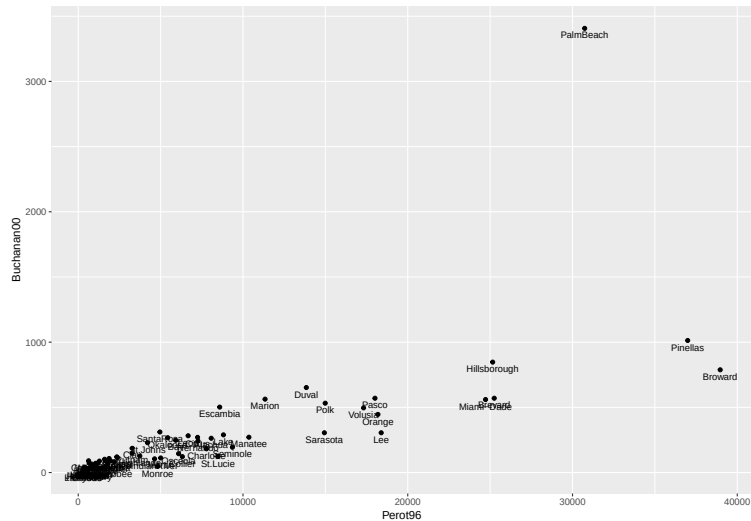
Plot votes for Perot and votes for Buchanan



Add label text of county

```
ggplot(florida, mapping=aes(x=Perot96, y=Buchanan00,  
  label=county)) +  
  geom_point() +  
  geom_text(vjust=1.5, size=3)
```

Add label text of county



Review: Linear Regression

We can write the linear regression model as:

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \varepsilon_i$$

- ▶ Y_i : dependent variable
- ▶ X_i : independent variable
- ▶ β_0 : intercept
- ▶ β_1 : slope
- ▶ ε_i : error

Review: Linear Regression

We fit the line by minimizing the sum of squared errors:

- Find $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimizes the sum of the squared errors, $\sum_{i=1}^N (Y_i - \hat{Y}_i)^2$

Interpretation:

- A one unit increase in X is associated with $\hat{\beta}_1$ -unit change in Y .

Prediction Buchanan00

- ▶ Let's predict the number of votes in a county for Buchanan in 2000, based on the number of votes for Perot in 1996.

$$Buchanan00_i = \beta_0 + \beta_1 \cdot Perot96_i + \varepsilon_i$$

- ▶ Then we can form predicted values for each county
- ▶ If the predicted value for Palm Beach is far from the actual value, then something unusual is happening in Palm Beach

lm(function)

- ▶ The function you use to estimate linear regression in R is the `lm()` function (which stands for linear model). The general syntax is

```
lm.fit <- lm(yvar ~ xvar, data=df)
```

- ▶ `lm.fit` stores the estimates of the linear regression as an object in R
- ▶ `yvar` should be replaced with the **dependent** variable
- ▶ `xvar` should be replaced with the **independent** variable
- ▶ `df` should be replaced with whatever the dataframe is named in your environment

Estimate regression predicting Buchanan00 with Perot96

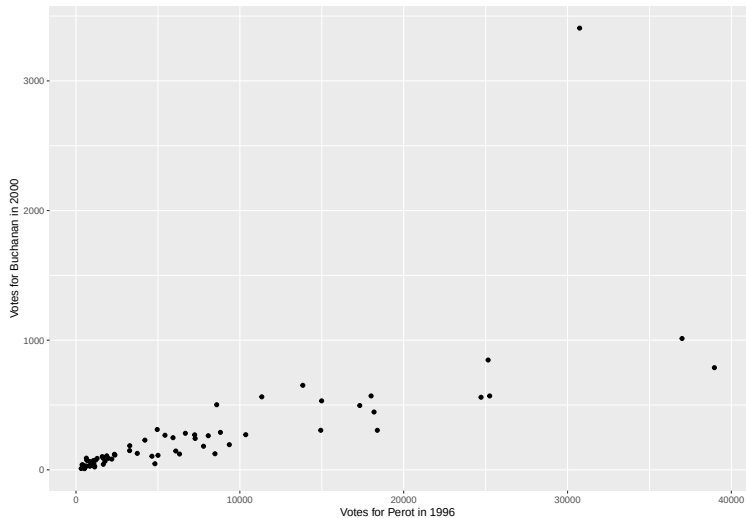
```
lm.florida <- lm(Buchanan00 ~ Perot96, data=florida)
lm.florida

##
## Call:
## lm(formula = Buchanan00 ~ Perot96, data = florida)
##
## Coefficients:
## (Intercept)      Perot96
##      1.34575      0.03592
```

Interpretation

- ▶ We found $\hat{\beta}_1 = 0.036$
 - ▶ Implies an additional vote for Perot in 1996 is predicted to increase votes for Buchanan by 0.036
- ▶ Now let's learn how to add a linear regression line to our scatterplot

Plot votes for Perot and votes for Buchanan in ggplot



geom_smooth

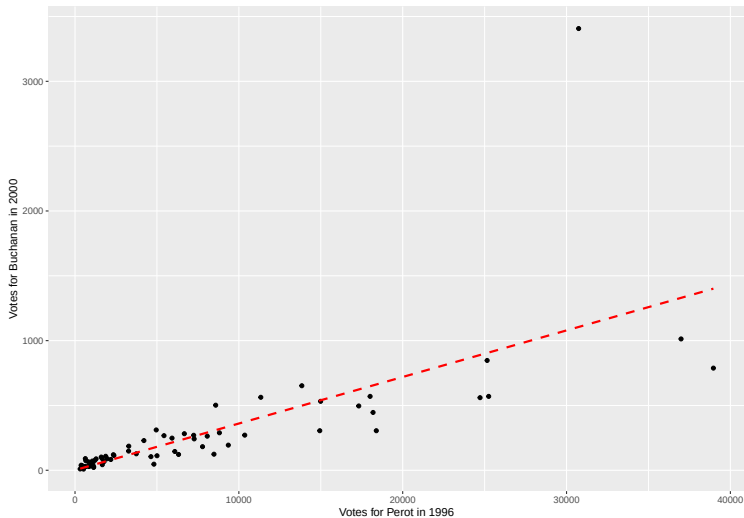
- ▶ To add a regression line we can use the `geom_smooth` option in `ggplot`
- ▶ To add a linear regression line we specify `method="lm"` where `lm` stands for linear model
- ▶ Where are also going to set `se=FALSE` (which will prevent confidence intervals from being displayed) and change the color of the regression line to red
- ▶ We can also specify the `linetype` in `geom_smooth` (i.e. solid or dashed)

Plot the regression line

```
ggplot(florida, mapping=aes(x=Perot96, y=Buchanan00)) +  
  geom_point() +  
  xlab("Votes for Perot in 1996") +  
  ylab("Votes for Buchanan in 2000") +  
  geom_smooth(method = "lm", se = FALSE, color="red", linetype="dashed")
```

Plot the regression line

```
## `geom_smooth()` using formula = 'y ~ x'
```



Review: Predicted Values

Predicted values (sometimes called “fitted values”): linear regression’s estimate of what Y should be for a given X

$$\blacktriangleright \hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 * X_i$$

In our example:

$$\blacktriangleright \text{Buchanan00}_i = \hat{\beta}_0 + \hat{\beta}_1 * \text{Perot96}_i$$

- $\blacktriangleright$ Palm Beach County had 30739 votes for Perot in 1996. The linear regression would predict

$$\text{Predicted Buchanan} = 1.346 + 30739 \cdot 0.036 \approx 1108$$

Review: Residuals

Residual: The difference between the actual and predicted

- ▶ $Y_i - \hat{Y}_i$

In our example:

- ▶ $Buchanan00_i - \hat{Buchanan00}_i$

In R we can add the residual for each county to the data frame

- ▶ This will allow us to see if Palm Beach's error is unusually large in magnitude

Estimate regression predicting Buchanan00 with Perot96

```
lm.florida <- lm(Buchanan00 ~ Perot96, data=florida)
lm.florida

##
## Call:
## lm(formula = Buchanan00 ~ Perot96, data = florida)
##
## Coefficients:
## (Intercept)      Perot96
##      1.34575      0.03592
```

Use names to find saved output

```
lm.florida <- lm(Buchanan00 ~ Perot96, data=florida)
names(lm.florida)
```

```
## [1] "coefficients" "residuals"      "effects"         "rank"
## [5] "fitted.values" "assign"          "qr"              "df.residual"
## [9] "xlevels"       "call"           "terms"           "model"
```


Extract fitted values

```
florida$predBuchanan00 <- lm.florida$fitted.values
```

- ▶ This will add a variable to the data frame `florida` that stores the predicted values

Extract residuals

```
florida$residuals <- lm.florida$residuals
```

- ▶ This will add a variable to the data frame `florida` that stores the residuals

Order by largest residuals

```
arrange(florida, desc(abs(residuals)))
```

##	county	Clinton96	Dole96	Perot96	Bush00	Gore00	Buchanan00
## 1	PalmBeach	230621	133762	30739	152846	268945	3407
## 2	Broward	320736	142834	38964	177323	386561	788
## 3	Lee	65692	80882	18389	106141	73560	305
## 4	Brevard	80416	87980	25249	115185	97318	570
## 5	Miami-Dade	317378	209634	24722	289492	328764	560
## 6	Pinellas	184728	152125	36990	184823	200629	1013
## 7	Sarasota	63648	69198	14939	83100	72853	305
## 8	Orange	105513	106026	18191	134517	140220	446
## 9	Escambia	37768	60839	8587	73017	40943	502
## 10	St.Lucie	36168	28892	8482	34705	41559	124
## 11	Marion	37033	41397	11340	55141	44665	563
## 12	Duval	112258	126857	13844	152098	107864	652
## 13	Seminole	45051	59778	9357	75677	59174	194
## 14	SantaRosa	10923	26244	4957	36274	12802	311

Use pipe operator to do this

```
florida %>%  
  mutate(predBuchanan00 = lm.florida$fitted.values,  
         residuals=lm.florida$residuals) %>%  
  select(county, predBuchanan00, Buchanan00,residuals) %>%  
  arrange(desc(abs(residuals)))
```

##	county	predBuchanan00	Buchanan00	residuals
## 1	PalmBeach	1105.33818	3407	2.301662e+03
## 2	Broward	1400.73939	788	-6.127394e+02
## 3	Lee	661.78743	305	-3.567874e+02
## 4	Brevard	908.16461	570	-3.381646e+02
## 5	Miami-Dade	889.23738	560	-3.292374e+02
## 6	Pinellas	1329.84310	1013	-3.168431e+02
## 7	Sarasota	537.88054	305	-2.328805e+02
## 8	Orange	654.67625	446	-2.086763e+02
## 9	Escambia	309.74820	502	1.922518e+02
## 10	St.Lucie	305.97713	124	-1.819771e+02

Conclusion

- ▶ The residual is largest in magnitude in Palm Beach (around 2,300 “extra” votes for Buchanan)
 - ▶ The next largest residual in magnitude is around 612 votes
- ▶ Remember: Bush only won Florida 537 votes. Quite possible that the butterfly ballot in Palm Beach altered the outcome of the election!